

Java Lab Report: Student Utility Program

Object-Oriented Programming with Java

Language: Java (JDK 8+)

Paradigm: OOP / Console Application

Document Type: Technical Documentation

1. Objective

To design, develop, and execute an interactive menu-driven Java console application utilizing Object-Oriented Programming (OOP) concepts. The program manages student academic details, including entering marks for three subjects, calculating total marks and percentage, and displaying structured student records.

2. System Requirements

Hardware Requirements

Component	Minimum Requirement	Recommended Specification
Processor	1.6 GHz Dual-Core CPU	Intel Core i5 / AMD Ryzen 5 or higher
RAM	2 GB	8 GB or higher
Hard Disk Space	500 MB (for JDK and execution)	2 GB free storage space
Display	1024 x 768 resolution	1920 x 1080 FHD

Software Requirements

Software Component	Specification / Version
Operating System	Windows 10/11, macOS, or Linux (Ubuntu/Debian)
Java Development Kit (JDK)	JDK 8 / JDK 11 / JDK 17 / JDK 21 or higher
Compiler & Runtime	javac (Java Compiler) & java (Java Virtual Machine)
IDE / Text Editor	VS Code, Eclipse, IntelliJ IDEA, or Notepad++

3. Algorithm

The system execution is divided into class operations and main utility control flow:

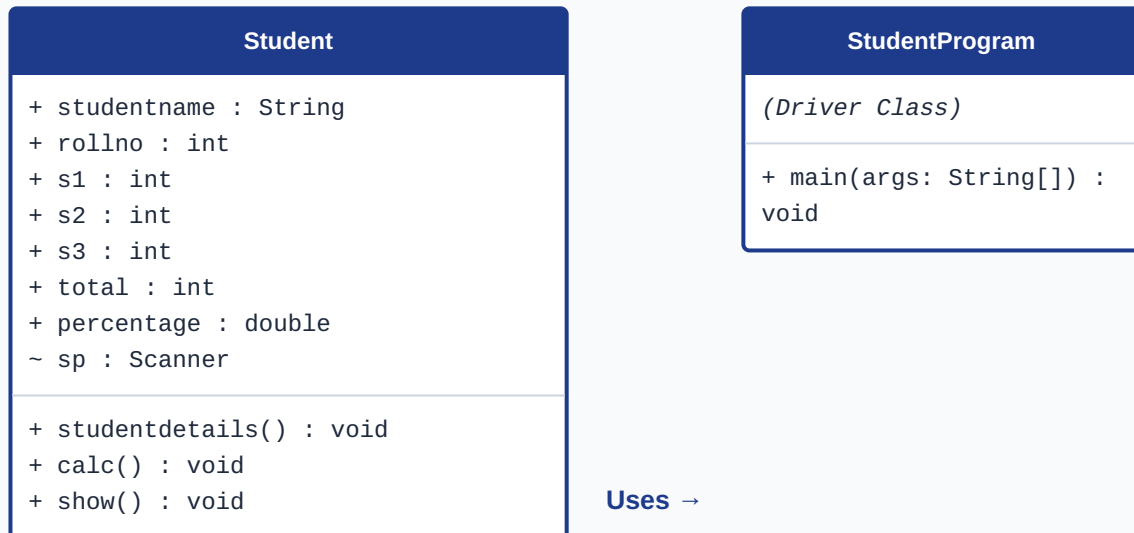
Main Control Flow Algorithm

1. **Start** the program.
2. Instantiate the `Scanner` object for main menu input and create an instance of `Student` class named `st`.
3. Display the menu options:
 - 1: Enter Student Details
 - 2: Calculate Total and Percentage
 - 3: Show Result
 - 4: Exit
4. Read the user's menu choice.
5. Perform operation using `switch(choice)`:
 - **Case 1:** Call `st.studentdetails()`. Prompt user for student name, roll number, and subject marks (`s1`, `s2`, `s3`). Clear input buffer.
 - **Case 2:** Call `st.calc()`. Compute $total = s1 + s2 + s3$ and $percentage = total / 3.00$. Display calculation result.
 - **Case 3:** Call `st.show()`. Print all stored details in formatted output.
 - **Case 4:** Display exit message and terminate loop.
 - **Default:** Print "Invalid choice" prompt.
6. Repeat steps 3–5 while `choice != 4`.
7. Close scanner objects and **Stop** program execution.

4. Class & UML Diagrams

Class Diagram (UML Representation)

The program consists of two main classes: Student (data model & operations) and StudentProgram (driver application).



5. Source Code

```
import java.util.Scanner;

class Student {
    String studentname;
    int rollno;
    int s1, s2, s3;
    int total;
    double percentage;
    Scanner sp = new Scanner(System.in);

    void studentdetails() {
        System.out.println("enter the name of student");
        studentname = sp.nextLine();
        System.out.println("enter th roll number");
        rollno = sp.nextInt();
        System.out.println("enter the firstsubject marks");
        s1 = sp.nextInt();
        System.out.println("enter the secondsubject marks");
        s2 = sp.nextInt();
        System.out.println("enter the thirdsubject marks");
        s3 = sp.nextInt();
        sp.nextLine(); // Clear buffer
    }

    // total and percentage
    void calc() {
```

```

        total = s1 + s2 + s3;
        System.out.println("total is " + total);
        percentage = total / 3.00;
        System.out.println("percentage is " + percentage);
        System.out.println("the caluculation is completed");
    }

    void show() {
        System.out.println("----student details ----");
        System.out.println(" name of student is " + studentname);
        System.out.println(" roll number is " + rollno);
        System.out.println(" s1 marks is " + s1);
        System.out.println(" s2 marks is " + s2);
        System.out.println(" s3 marks is " + s3);
        System.out.println(" total marks is " + total);
        System.out.println(" total percentage is " + percentage);
    }
}

public class StudentProgram {
    public static void main(String[] args) {
        Scanner sp = new Scanner(System.in);
        Student st = new Student();
        int choice;

        do {
            System.out.println("--- student utuility program---");
            System.out.println("1.enter student details ");
            System.out.println("2.calc total and percentage");
            System.out.println("3.show result");
            System.out.println("4.exit");
            System.out.println("enter your choice");
            choice = sp.nextInt();

            switch (choice) {
                case 1:
                    st.studentdetails();
                    break;
                case 2:
                    st.calc();
                    break;
                case 3:
                    st.show();
                    break;
                case 4:
                    System.out.println("exited ");
                    break;
                default:
                    System.out.println("invalid choice");
            }
        } while (choice != 4);

        sp.close();
    }
}

```

6. Compilation & Execution Steps

Step 1: Save the File

Save the Java code in a file named `StudentProgram.java` (matching the public class name).

Step 2: Open Terminal / Command Prompt

Navigate to the directory where `StudentProgram.java` is stored.

Step 3: Compile the Program

Run the Java compiler command:

```
javac StudentProgram.java
```

Note: If no errors are displayed, two bytecode files will be generated: `Student.class` and `StudentProgram.class`.

Step 4: Execute the Application

Run the compiled class using the Java Virtual Machine:

```
java StudentProgram
```

7. Program Sample Output

Below is the console output from executing the program complete with user interactions:

```
--- student utuility program---
1.enter student details
2.calc total and percentage
3.show result
4.exit
enter your choice
1
enter the name of student
John Doe
enter th roll number
101
enter the firstsubject marks
85
```

```
enter the secondsubject marks
90
enter the thirdsubject marks
88

--- student utuility program----
1.enter student details
2.calc total and percentage
3.show result
4.exit
enter your choice
2
total is 263
percentage is 87.66666666666667
the caluculation is completed

--- student utuility program----
1.enter student details
2.calc total and percentage
3.show result
4.exit
enter your choice
3
-----student details -----
name of student is John Doe
roll number is 101
s1 marks is 85
s2 marks is 90
s3 marks is 88
total marks is 263
total percentage is 87.66666666666667

--- student utuility program----
1.enter student details
2.calc total and percentage
3.show result
4.exit
enter your choice
4
exited
```